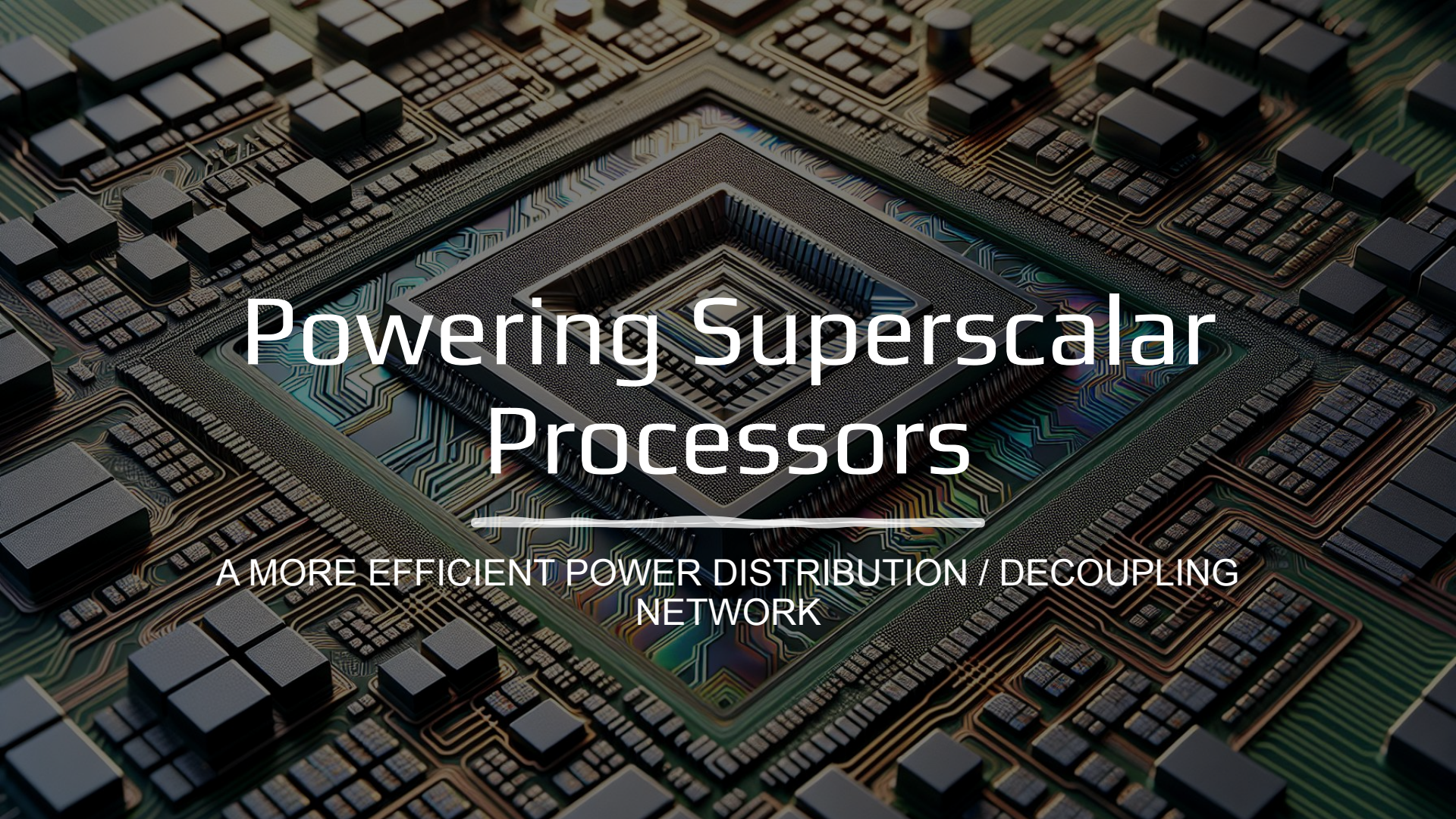




Powering Superscalar Processors

A MORE EFFICIENT POWER DISTRIBUTION / DECOUPLING
NETWORK



ISSUES FACED

Main problem in modern processors	Why current solutions are inefficient
1. Huge current demand	Low core voltage requires hundreds of amps. Delivering this current causes losses throughout the VRM, package and PDN.
2. Fast current transients	VRM inductors cannot change current instantly, so capacitors must temporarily supply the difference. This requires substantial decoupling infrastructure.
3. Voltage droop	PDN resistance and inductance cause . Existing systems mainly react to the voltage disturbance rather than preventing the demand.
4. Thermal limits	DVFS/frequency throttling reduces power by slowing large portions of the processor, even when only certain execution units are responsible for the power increase.
5. Inefficient power allocation	Conventional power management generally treats power as a domain/core-level resource rather than deciding which individual instructions are worth spending power on.
6. Local power hotspots	Global temperature/power limits can throttle an entire core/domain even when only one region is creating the hotspot.

Machine Learning-Based Microarchitecture-Level Power Modeling of CPUs

Ajay Krishna Ananda Kumar, Sami Alsalamini,
Hussam Amrouh, *Member, IEEE*, and Andreas Gerstlauer, *Senior Member, IEEE*

Abstract—Energy efficiency has emerged as a key concern for modern processor design, especially when it comes to embedded and mobile devices. It is vital to accurately quantify the power consumption of different micro-architectural components in a CPU. Traditional RTL or gate-level power estimation is too slow for early design-space exploration studies. By contrast, existing architecture-level power models suffer from large inaccuracies. Recently, advanced machine learning techniques have been demonstrated for fixed-function accelerators and simple in-order cores with predicted machine learning-based approach for microarchitecture-level power modeling of activity traces obtained from microarchitecture simulations. We extract representative formulations for different types of CPU-internal structures. Cycle-accurate models, number of gate-level simulations and hierarchically composed to build power model in-order and out-of-order RISC-V cores. Cross-validation results show that our model is 3% of a gate-level power estimation on average. In addition, our power model for the micro-benchmarks can predict the cycle-by-cycle power of real-world applications.

Index Terms—Machine learning, power modeling, micro-architecture simulation

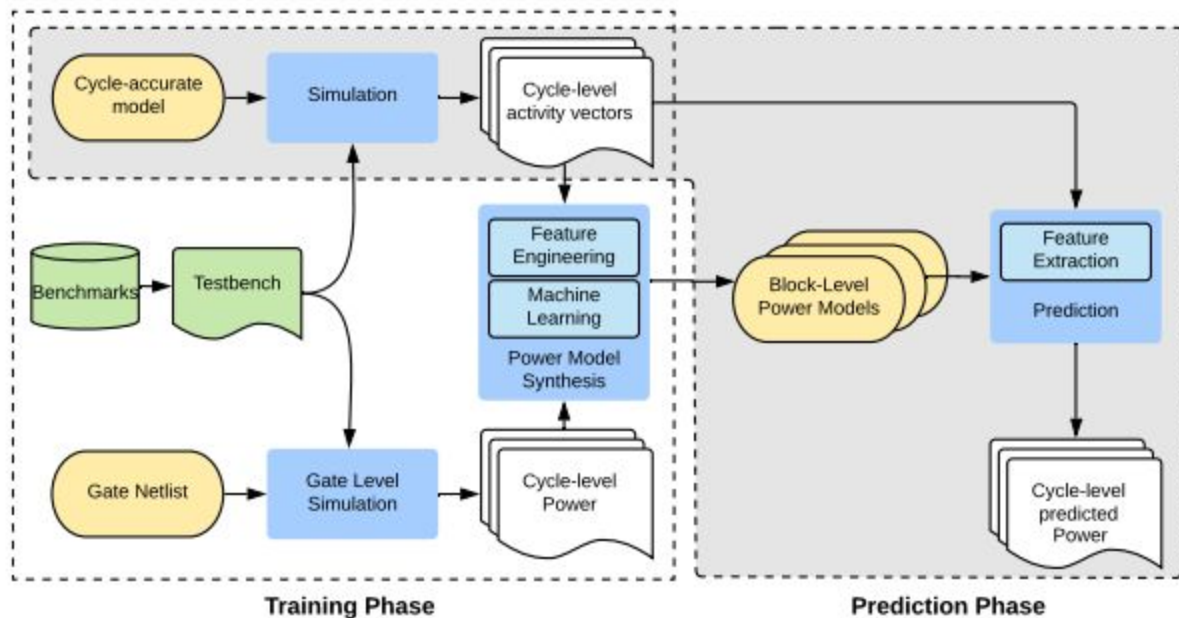


Fig. 2. Power modeling flow.

An Improved Active Decoupling Capacitor for “Hot-Spot” Supply Noise Reduction in ASIC Designs

Xiongfei Meng, *Student Member, IEEE*, and Resve Saleh, *Fellow, IEEE*

Abstract—On-chip decoupling capacitors (decaps) are widely used to reduce power supply noise by placing them at the appropriate locations on the chip between blocks. While passive decaps can provide a certain degree of protection against IR drop, if a problem is found after the physical design is completed, it is difficult to implement a quick fix to the problem. In this paper, we investigate the use of an active decap as a drop-in replacement for passive decaps to provide noise reduction for these so-called “hot-spot” IR drop problems found late in the design process. A modified active decap design is proposed for ASIC applications operating up to 1 GHz. Our improvement uses latch-based comparators as the sensing circuit, which provides a better power/delay tradeoff than previous designs and incorporates hysteresis to minimize unnecessary switching. It is implemented in a 1 V-core 90 nm CMOS process with a total area of 0.085 mm² and static power of 2.8 mW. Measurements from a number of test chips show that using an active decap can provide between 10%–20% noise reduction in the 200 MHz–1 GHz frequency range over its passive counterpart. Sizing and placement analyses are also carried out using circuit simulation. The active decap is most effective when placed in close proximity to the hot-spot, as compared to the passive decap which is less sensitive to the exact location. Overall, if sized and placed properly, active decaps can provide an additional 20% reduction in supply noise over passive decaps.

Index Terms—MOS integrated circuits, decoupling capacitors, comparators, power supply noise.

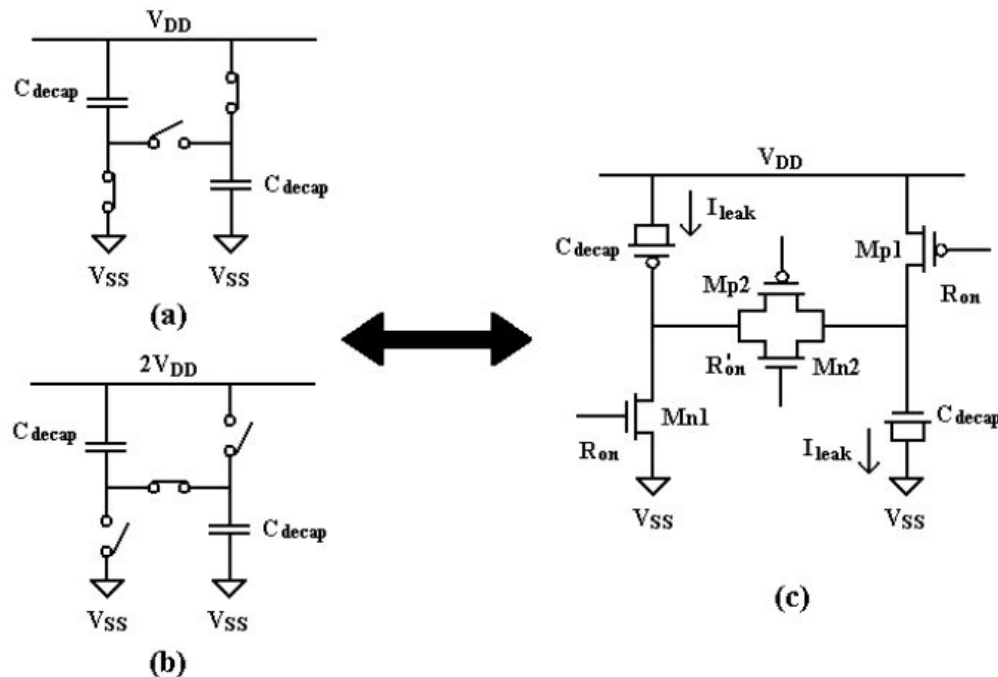


Fig. 1. Active decap concept and its MOS implementation. (a) Decaps in parallel. (b) Decaps in series. (c) Circuit implementation.

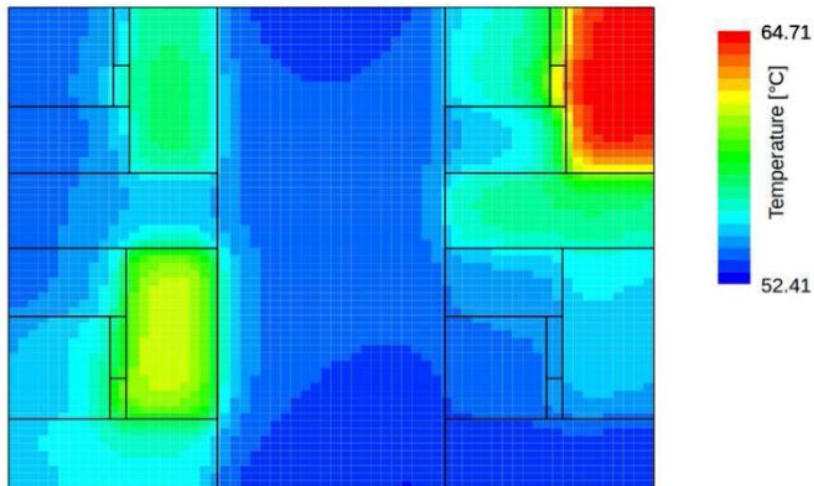
HotSniper

Anuj Pathania

Sep 8, 2021

PDF

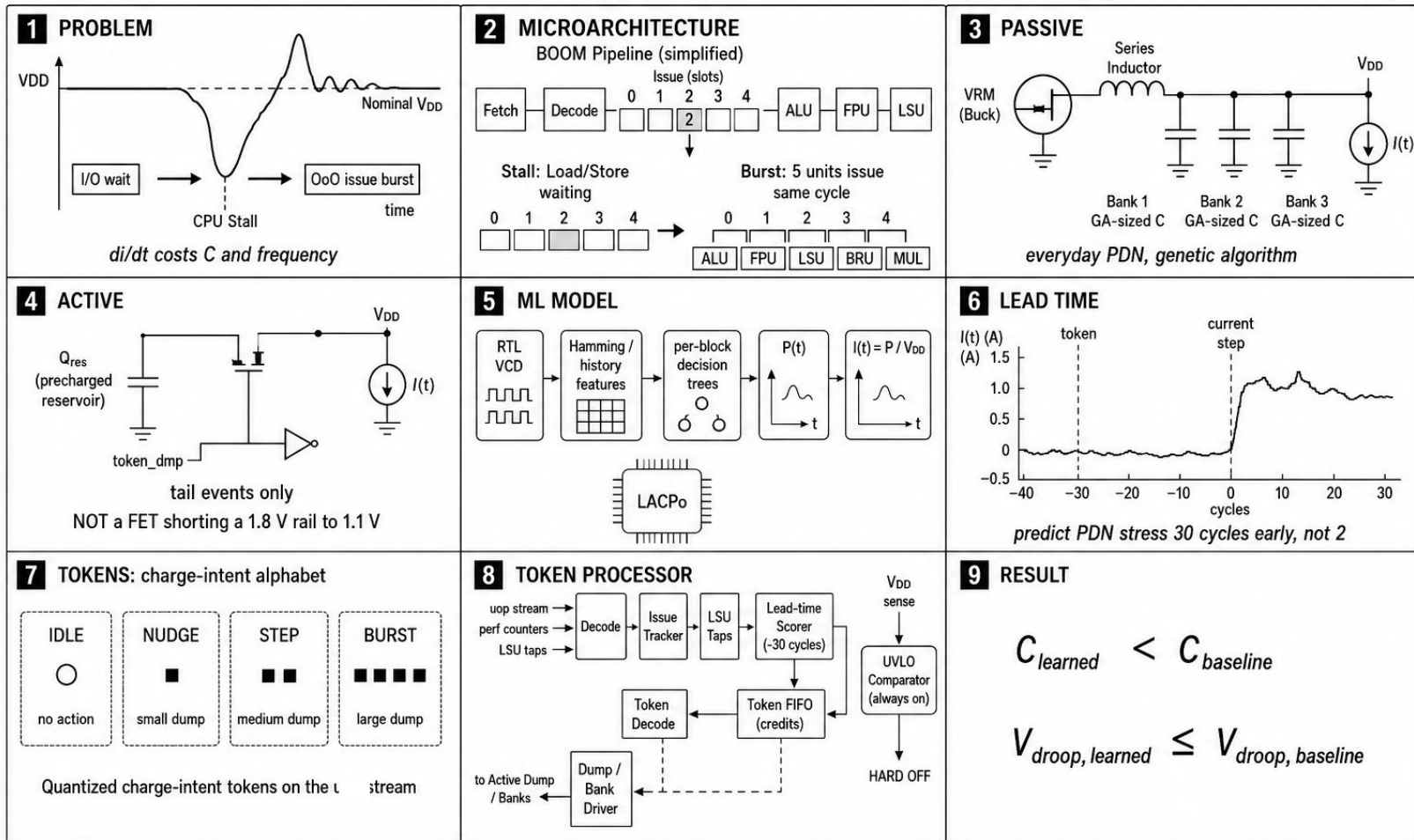
Code



HotSniper is an EDA toolchain for interval thermal simulations of 2D multi-/many-cores in an open system.

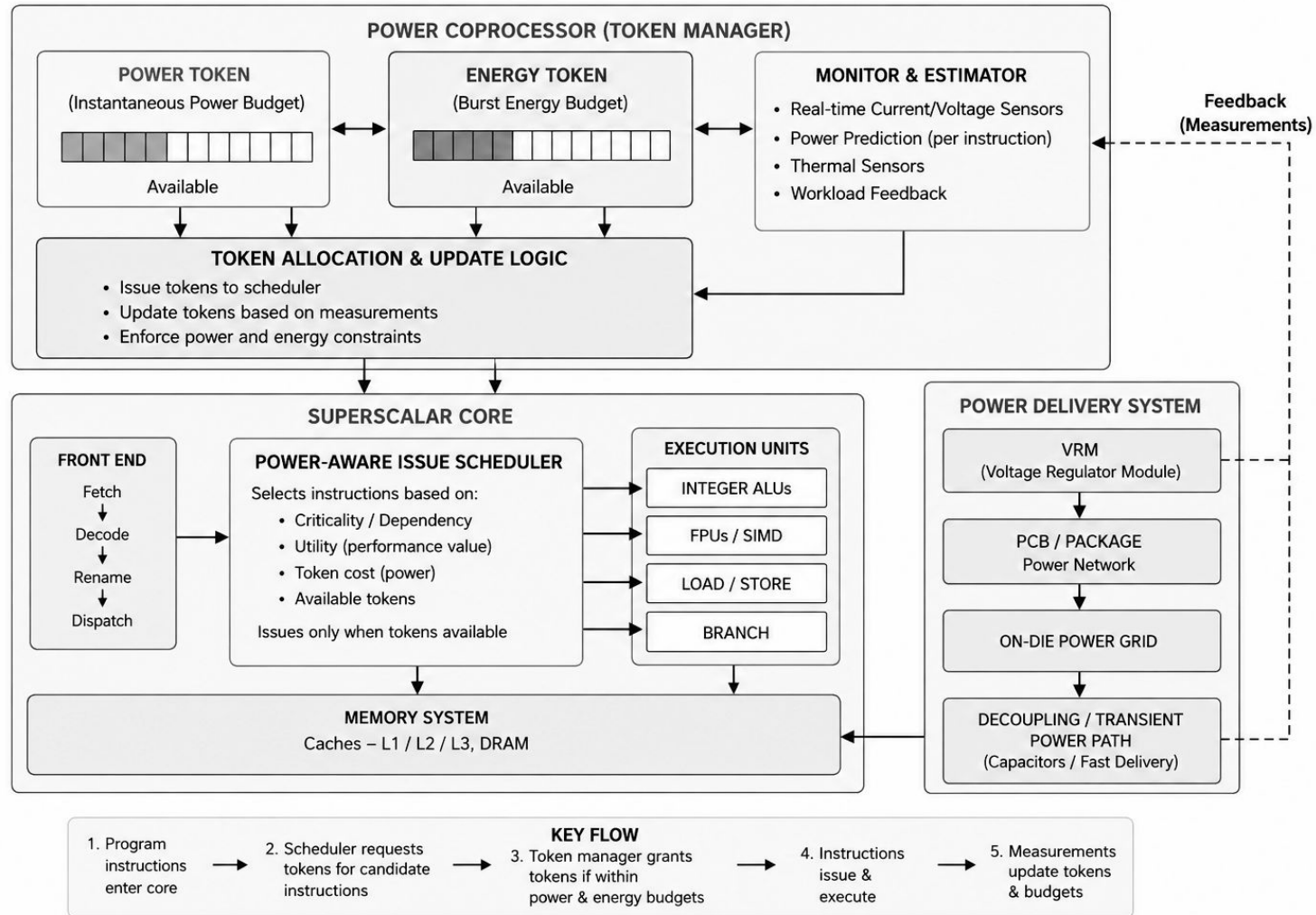
Simulator

Instruction-Timed Power Delivery: GA Passives + Token-Triggered Active Dump



Passive = typical $I(t)$. Active = rare worst case. Tokens = feedforward. Safety envelope always on.

POWER-AWARE SUPERSCALAR PROCESSOR ARCHITECTURE



Dummy-VCD programs through the same LACPo models



Early microarch signals vs same-cycle power nets

Tokens need ~30 clocks of warning. LACPo nets estimate I(t) now.



Early enough — use as tokens

Stall falling / I-cache miss / fetch resume

20-40+ clk stall-then-burst

D-cache miss / MSHR allocate

10-40+ clk wakeup after fill

Fetch + decode opcode (int / mem / FP)

8-15 clk which units will fire

ROB empty → full, dispatch valids

10-20 clk OoO issue ramp

Too late — LACPo only

ALU / FPU operand Hamming

same cycle as execute current

Integer / FP register-file access

same cycle as RF current

LSU memreq / exe_resp

same cycle as memory current

Issue fire, commit, writeback

0-2 clk cannot arm a dump

$30 \text{ clk} \times 3 \text{ ns} \approx 90 \text{ ns}$ to pre-enable the dump. Pipeline depth alone is only 8-15 clk; the extra lead is stall, miss, and ROB fill.

Medium BOOM · schematic pipeline/event latency, not a measured VCD study